## A7

## Q1

### a) $S = \{s_1 + s_2 : s_1 \in S_1, s_2 \in S_2\}$, where $S_1$ and $S_2$ are convex: Yes

Take two points $a$ and $b$ in $S$. Since $a \in S$, there exist $s_1^a \in S_1$ and $s_2^a \in S_2$ such that $a = s_1^a + s_2^a$. Similarly, $b = s_1^b + s_2^b$ with $s_1^b \in S_1$ and $s_2^b \in S_2$.

Consider the convex combination:

$$\lambda a + (1-\lambda)b = \lambda(s_1^a + s_2^a) + (1-\lambda)(s_1^b + s_2^b) = [\lambda s_1^a + (1-\lambda)s_1^b] + [\lambda s_2^a + (1-\lambda)s_2^b].$$

- Since $S_1$ is convex and $s_1^a, s_1^b \in S_1$, it follows that $\lambda s_1^a + (1-\lambda)s_1^b \in S_1$.
- Similarly, since $S_2$ is convex, $\lambda s_2^a + (1-\lambda)s_2^b \in S_2$.

Thus, $[\lambda s_1^a + (1-\lambda)s_1^b] + [\lambda s_2^a + (1-\lambda)s_2^b]$ is the sum of an element from $S_1$ and an element from $S_2$, which is in $S$ by definition. Hence, $S$ is convex.

### b) $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 5\}$: No

This set represents a circle in $\mathbb{R}^2$ with radius $\sqrt{5}$ centered at the origin, as $x^2 + y^2 = 5$.
Select two points on the circle, such as $(0, \sqrt{5})$ and $(0, -\sqrt{5})$:

- $(0, \sqrt{5})$: $0^2 + (\sqrt{5})^2 = 5$: in the set.
- $(0, -\sqrt{5})$: $0^2 + (-\sqrt{5})^2 = 5$: in the set.

The line segment between them is parametrized as $(0, \lambda\sqrt{5} + (1-\lambda)(-\sqrt{5})) = (0, (2\lambda - 1)\sqrt{5})$ for $\lambda \in [0, 1]$. Compute $x^2 + y^2$ for a point on this segment, say at $\lambda = 0.5$:

- Point: $(0, 0)$, since $(2 \cdot 0.5 - 1)\sqrt{5} = 0$.
- $x^2 + y^2 = 0^2 + 0^2 = 0 < 5$: not in the set.

Since the midpoint (and indeed all interior points between the endpoints) does not satisfy $x^2 + y^2 = 5$, the set does not contain the entire line segment between two of its points.

### c) $\{x \in \mathbb{R} : ax^2 + 2x - \frac{1}{a} \geq 0\}$ with $a > 0$: No

Define $f(x) = ax^2 + 2x - \frac{1}{a}$, and the set is $\{x : f(x) \geq 0\}$.
Since $a > 0$, $f(x)$ is a quadratic function with a parabola opening upwards.

Find the roots of $f(x) = 0$:

$$ax^2 + 2x - \frac{1}{a} = 0$$

Using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ with coefficients $a = a$, $b = 2$, $c = -\frac{1}{a}$:

$$\Delta = 2^2 - 4 \cdot a \cdot \left(-\frac{1}{a}\right) = 4 + 4 = 8,$$

$$x = \frac{-2 \pm \sqrt{8}}{2a} = \frac{-2 \pm 2\sqrt{2}}{2a} = \frac{-1 \pm \sqrt{2}}{a}.$$

Roots are:

- $x_1 = \frac{-1-\sqrt{2}}{a}$,
- $x_2 = \frac{-1+\sqrt{2}}{a}$.

Since $a > 0$ and $-1 - \sqrt{2} < -1 + \sqrt{2}$, $x_1 < x_2$. Because the parabola opens upwards, $f(x) \geq 0$ when $x \leq x_1$ or $x \geq x_2$. Thus, the set is:

$$(-\infty, x_1] \cup [x_2, \infty).$$

Take $p = x_1 - 1 \in (-\infty, x_1]$ and $q = x_2 + 1 \in [x_2, \infty)$. Their midpoint is:

$$\frac{p+q}{2} = \frac{(x_1 - 1) + (x_2 + 1)}{2} = \frac{x_1 + x_2}{2} = \frac{\frac{-1-\sqrt{2}}{a} + \frac{-1+\sqrt{2}}{a}}{2} = -\frac{1}{a}.$$

Since $x_1 < -\frac{1}{a} < x_2$ (as $-1 - \sqrt{2} < -1 < -1 + \sqrt{2}$), evaluate $f$ at $x = -\frac{1}{a}$:

$$f\left(-\frac{1}{a}\right) = a\left(-\frac{1}{a}\right)^2 + 2\left(-\frac{1}{a}\right) - \frac{1}{a} = a \cdot \frac{1}{a^2} - \frac{2}{a} - \frac{1}{a} = \frac{1}{a} - \frac{3}{a} = -\frac{2}{a} < 0,$$

since $a > 0$. Thus, $-\frac{1}{a}$ is not in the set, so the set is not convex.

**d) $\{(x, y) \in \mathbb{R}^2 : y \geq |x|\}$: Yes**

This set includes all points $(x, y)$ where $y$ is at least the absolute value of $x$, forming a region above the V-shaped graph of $y = |x|$. Rewrite the condition $y \geq |x|$ as:

- $y \geq x$ (when $x \geq 0$),
- $y \geq -x$ (when $x < 0$),
  which holds if $y \geq x$ and $y \geq -x$ for all $x$. Thus, the set is:

$$\{(x, y) : y \geq x\} \cap \{(x, y) : y \geq -x\}.$$

  Each set is a half-plane:
- $y \geq x$ is the region above the line $y = x$,
- $y \geq -x$ is the region above the line $y = -x$.

Half-planes are convex, and the intersection of convex sets is convex.
Take two points $(x_1, y_1)$ and $(x_2, y_2)$ where $y_1 \geq |x_1|$ and $y_2 \geq |x_2|$. Their convex combination is:

$$(\lambda x_1 + (1 - \lambda)x_2, \lambda y_1 + (1 - \lambda)y_2).$$

The following is in need:

$$\lambda y_1 + (1 - \lambda)y_2 \geq |\lambda x_1 + (1 - \lambda)x_2|.$$

Since $|\cdot|$ is convex, $|\lambda x_1 + (1 - \lambda)x_2| \leq \lambda|x_1| + (1 - \lambda)|x_2|$. Given $y_1 \geq |x_1|$ and $y_2 \geq |x_2|$:

$$\lambda y_1 + (1 - \lambda)y_2 \geq \lambda|x_1| + (1 - \lambda)|x_2| \geq |\lambda x_1 + (1 - \lambda)x_2|,$$

so the combination is in the set.

**e) $\{x \in \mathbb{R}^2 : x^\top Ax \leq 0\}$ where $A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$: No**

For $x = (x_1, x_2)$, compute:

$$x^\top Ax = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} x_1 \\ -x_2 \end{pmatrix} = x_1^2 - x_2^2.$$

The set is:

$$\{(x_1, x_2) : x_1^2 - x_2^2 \leq 0\} \Leftrightarrow \{(x_1, x_2) : |x_2| \geq |x_1|\}\}.$$

This includes points where $x_2 \geq |x_1|$ (upper region) or $x_2 \leq -|x_1|$ (lower region). Test with points $(1, 2)$ and $(1, -2)$:

- $(1, 2)$: $|2| = 2 \geq 1$, or $2^2 - 1^2 = 4 - 1 = 3 \geq 0$, yes.
- $(1, -2)$: $|-2| = 2 \geq 1$, or $(-2)^2 - 1^2 = 4 - 1 = 3 \geq 0$, yes.

The line segment is $(1, 2(1 - \lambda) + (-2)\lambda) = (1, 2 - 4\lambda)$ for $\lambda \in [0, 1]$. At $\lambda = 0.5$:

- Point: $(1, 0)$,
- $0^2 - 1^2 = -1 < 0$ or $|0| < |1|$, not in the set.

Since $(1, 0)$ lies between $(1, 2)$ and $(1, -2)$ but is not in the set, it's not convex. Alternatively, $A$ is indefinite (eigenvalues 1 and -1), and $\{x : x^\top Ax \leq 0\}$ is convex only if $A$ is negative semidefinite, which it is not.

## Q2

**Part (a): $f(x) = \|x\|_1 = \sum_{i=1}^d |x_i|$: Yes**

Consider two vectors $x, y \in \mathbb{R}^d$ and $\lambda \in [0, 1]$. Now need to verify:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

LHS:

$$f(\lambda x + (1 - \lambda)y) = \sum_{i=1}^d |\lambda x_i + (1 - \lambda)y_i|$$

By the triangle inequality for real numbers, $|a + b| \leq |a| + |b|$, and since $\lambda \geq 0$ and $1 - \lambda \geq 0$:

$$|\lambda x_i + (1 - \lambda)y_i| \leq \lambda|x_i| + (1 - \lambda)|y_i|$$

Summing over all components:

$$\sum_{i=1}^{d} |\lambda x_i + (1-\lambda)y_i| \leq \sum_{i=1}^{d} [\lambda|x_i| + (1-\lambda)|y_i|] = \lambda \sum_{i=1}^{d} |x_i| + (1-\lambda) \sum_{i=1}^{d} |y_i| = \lambda f(x) + (1-\lambda) f(y)$$

Thus, the convexity inequality holds, so $f(x) = \|x\|_1$ is convex.

**Part (b): $f(x) = \|x - y\|^2$, where $y$ is a given vector in $\mathbb{R}^d$: Yes**

Since it's twice differentiable: examine the Hessian.

$$\frac{\partial f}{\partial x_i} = 2(x_i - y_i)$$

$$\frac{\partial^2 f}{\partial x_i^2} = 2, \quad \frac{\partial^2 f}{\partial x_i \partial x_j} = 0 \text{ for } i \neq j$$

The Hessian is a $d \times d$ matrix with 2 on the diagonal and 0 elsewhere, i.e., $2I$, where $I$ is the identity matrix. Since all eigenvalues of $2I$ are 2 (positive), the Hessian is positive definite, implying strict convexity.

**Part (c): $f : \mathbb{R}^d \to \mathbb{R}$ with $f(\lambda x) = |\lambda| f(x)$ and $f(x + y) \leq f(x) + f(y)$: Yes**

This function satisfies two properties: homogeneity of degree 1 ($f(\lambda x) = |\lambda| f(x)$ for all $\lambda \in \mathbb{R}, x \in \mathbb{R}^d$) and subadditivity ($f(x + y) \leq f(x) + f(y)$).

For $x, y \in \mathbb{R}^d$ and $\lambda \in [0, 1]$:

$$f(\lambda x + (1-\lambda)y) \leq f(\lambda x) + f((1-\lambda)y)$$

Using homogeneity, since $\lambda \geq 0$ and $1 - \lambda \geq 0$:

$$f(\lambda x) = |\lambda| f(x) = \lambda f(x), \quad f((1-\lambda)y) = |1-\lambda| f(y) = (1-\lambda) f(y)$$

Thus:

$$f(\lambda x + (1-\lambda)y) \leq \lambda f(x) + (1-\lambda) f(y)$$

This matches the convexity condition exactly. Hence, $f$ is convex.

**Part (d): $f(x) = |x|^{3/2}$: Yes**

Assume $x \in \mathbb{R}$ (since the notation suggests a scalar, and vector interpretation is unclear without a norm).
The function isn't twice differentiable at $x = 0$, so compute the first derivative and assess if it's increasing (for differentiable functions, $f'$ increasing implies convexity).
For $x > 0$, $f(x) = x^{3/2}$:

$$f'(x) = \frac{3}{2} x^{1/2}$$

For $x < 0$, $f(x) = (-x)^{3/2}$, and using the chain rule:

$$f'(x) = \frac{3}{2}(-x)^{1/2} \cdot (-1) = -\frac{3}{2}(-x)^{1/2}$$

At $x = 0$, $f'(0) = \lim_{h \to 0} \frac{|h|^{3/2}}{h} = \lim_{h \to 0} |h|^{1/2} \cdot \frac{h}{|h|}$, which is 0 from both sides. Thus, $f'(x) = \frac{3}{2}|x|^{1/2} \cdot \operatorname{sgn}(x)$, increasing from negative to positive across $\mathbb{R}$. Since $f$ is differentiable and $f'$ is increasing, $f(x) = |x|^{3/2}$ is convex. Alternatively, $|x|$ is convex, and $t^{3/2}$ (for $t \geq 0$) is convex and increasing, so their composition is convex.

**Part (e): $f(x) = \ln\left(x + \sqrt{1 + x^2}\right) + x^2$: Yes**

Assuming $x \in \mathbb{R}$ (consistent with the scalar notation), compute the second derivative.
Let $g(x) = \ln\left(x + \sqrt{1 + x^2}\right)$, so $f(x) = g(x) + x^2$.
For $g(x)$, set $h(x) = x + \sqrt{1 + x^2}$:

$$h'(x) = 1 + \frac{x}{\sqrt{1 + x^2}}, \quad g'(x) = \frac{h'(x)}{h(x)} = \frac{1 + \frac{x}{\sqrt{1+x^2}}}{x + \sqrt{1 + x^2}}$$

However, recognize that $h(x) = e^{\operatorname{arcsinh}(x)}$, so $g(x) = \operatorname{arcsinh}(x)$, and $g'(x) = \frac{1}{\sqrt{1+x^2}}$. Then:

$$g''(x) = -\frac{x}{(1 + x^2)^{3/2}}$$

For $f(x)$:

$$f'(x) = \frac{1}{\sqrt{1 + x^2}} + 2x, \quad f''(x) = -\frac{x}{(1 + x^2)^{3/2}} + 2$$

Since $f''(x)$ is even, test for $x \geq 0$:

$$f''(x) = 2 - \frac{x}{(1+x^2)^{3/2}}$$

The term $k(x) = \frac{x}{(1+x^2)^{3/2}}$ peaks at $x = \frac{1}{\sqrt{2}}$, with value $\frac{2}{3\sqrt{3}} < 2$, so $f''(x) > 0$ for all $x$. Thus, $f$ is strictly convex.

## Q3

### a) The union of a finite number of convex sets is always convex: **False**

Consider two convex sets in $\mathbb{R}^2$: $A = \{(0,0)\}$ and $B = \{(1,1)\}$, both singleton sets, which are convex since they contain only one point. Their union is $A \cup B = \{(0,0), (1,1)\}$. Now, take the points $(0,0)$ and $(1,1)$ from $A \cup B$. For $\lambda = 0.5$, the convex combination is:

$$\lambda(0,0) + (1-\lambda)(1,1) = 0.5(0,0) + 0.5(1,1) = (0.5, 0.5).$$

However, $(0.5, 0.5) \notin A \cup B$, since the union only includes $(0,0)$ and $(1,1)$. Thus, $A \cup B$ is not convex.

### b) If $f$ and $g$ are convex functions, then the composite function $h = f \circ g$, i.e., $h(x) = f(g(x))$, is always a convex function: **False**

Consider $g(x) = x^2 - 1$, which is convex since its second derivative $g''(x) = 2 > 0$.
Define $f(y) = |y|$, which is convex because it is the maximum of the convex functions $y$ and $-y$.
Then:

$$h(x) = f(g(x)) = |g(x)| = |x^2 - 1|.$$

Test convexity of $h$ with points $x = -1$ and $y = 1$:

- $g(-1) = (-1)^2 - 1 = 0$, so $h(-1) = |0| = 0$,
- $g(1) = 1^2 - 1 = 0$, so $h(1) = |0| = 0$,
- For $\lambda = 0.5$, $\lambda(-1) + (1-\lambda)1 = 0$, so $h(0) = |0^2 - 1| = |-1| = 1$,
- $\lambda h(-1) + (1-\lambda)h(1) = 0.5 \cdot 0 + 0.5 \cdot 0 = 0$.

Check the convexity condition:

$$h(0) = 1 > 0 = 0.5h(-1) + 0.5h(1).$$

Since $1 > 0$, the inequality is violated (convexity requires $\leq$), so $h$ is not convex.

Thus, $f$ and $g$ are convex, but $h = f \circ g$ is not, proving the statement is **false**.

### c) If $f$ and $g$ are convex functions, then the function $h = f \cdot g$, i.e., $h(x) = f(x) \cdot g(x)$, is always a convex function: **False**

Test with $f(x) = x^2$ and $g(x) = (x-1)^2$, both convex since $f''(x) = 2 > 0$ and $g''(x) = 2 > 0$. Compute:

$$h(x) = x^2(x-1)^2 = x^2(x^2 - 2x + 1) = x^4 - 2x^3 + x^2.$$

Check convexity via the second derivative:

- $h'(x) = 4x^3 - 6x^2 + 2x$,
- $h''(x) = 12x^2 - 12x + 2$.

Solve $h''(x) \geq 0$:

- Discriminant of $12x^2 - 12x + 2 = 0$: $(-12)^2 - 4 \cdot 12 \cdot 2 = 144 - 96 = 48 > 0$,
- Roots: $x = \frac{12 \pm \sqrt{48}}{24} = \frac{12 \pm 4\sqrt{3}}{24} = \frac{3 \pm \sqrt{3}}{6}$,
- Approximate roots: $x \approx 0.211$ and $x \approx 0.789$.

Since the quadratic opens upward (coefficient $12 > 0$), $h''(x) < 0$ between the roots (e.g., at $x = 0.5$, $h''(0.5) = 12(0.25) - 12(0.5) + 2 = 3 - 6 + 2 = -1 < 0$), so $h$ is not convex everywhere.

Thus, the statement is **false**.

### d) If $f$ is a convex function, then the set $S \equiv \{(y, \mathbf{x}) \in \mathbb{R} \times \mathbb{R}^n \mid y = f(\mathbf{x})\}$ is a convex set: **False**

Take $f(x) = x^2$ in $\mathbb{R}$ (so $n = 1$):

- $S = \{(y, x) \in \mathbb{R} \times \mathbb{R} \mid y = x^2\}$,
- Points: $(1,1)$ (since $f(1) = 1$) and $(-1,1)$ (since $f(-1) = 1$),
- Line segment with $\lambda = 0.5$: $(0.5 \cdot 1 + 0.5 \cdot 1, 0.5 \cdot 1 + 0.5 \cdot (-1)) = (1, 0)$,
- Check: $f(0) = 0^2 = 0 \neq 1$, so $(1,0) \notin S$.

Since the midpoint is not in $S$, the graph is not convex unless $f$ is linear (e.g., $f(x) = ax + b$, where $S$ is a hyperplane). For non-linear convex $f$, $S$ is not convex.

### e) A line segment in $\mathbb{R}^2$ is always a convex set: True

A line segment between points $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2$ is $\{\lambda\mathbf{a} + (1-\lambda)\mathbf{b} \mid \lambda \in [0,1]\}$. For any two points $\mathbf{p} = \lambda_1\mathbf{a} + (1-\lambda_1)\mathbf{b}$ and $\mathbf{q} = \lambda_2\mathbf{a} + (1-\lambda_2)\mathbf{b}$ ($\lambda_1, \lambda_2 \in [0,1]$), their convex combination is:

$$\mu\mathbf{p} + (1-\mu)\mathbf{q} = \mu[\lambda_1\mathbf{a} + (1-\lambda_1)\mathbf{b}] + (1-\mu)[\lambda_2\mathbf{a} + (1-\lambda_2)\mathbf{b}] = [\mu\lambda_1 + (1-\mu)\lambda_2]\mathbf{a} + [\mu(1-\lambda_1) + (1-\mu)(1-\lambda_2)]\mathbf{b}.$$

Define $\lambda' = \mu\lambda_1 + (1-\mu)\lambda_2$. Since $\mu, \lambda_1, \lambda_2 \in [0,1]$, $\lambda' \in [0,1]$, and the coefficients sum to 1, so this point is in the line segment. Thus, the set is convex.

## Q4

### Part (a): Prove that $\beta_0^* = 0$

The objective function is:

$$f(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^{n} \left(\beta_0 + \boldsymbol{\beta}^T \boldsymbol{x}_i - y_i\right)^2.$$

Take the partial derivative with respect to $\beta_0$, and at the minimizer $(\beta_0^*, \boldsymbol{\beta}^*)$, the partial derivative equals zero:

$$\frac{\partial f}{\partial \beta_0} = \sum_{i=1}^{n} 2\left(\beta_0 + \boldsymbol{\beta}^T \boldsymbol{x}_i - y_i\right) \cdot 1 = 2\sum_{i=1}^{n} \left(\beta_0 + \boldsymbol{\beta}^T \boldsymbol{x}_i - y_i\right) = 0.$$

Divide by 2, expand the sum:

$$\sum_{i=1}^{n} \beta_0^* + \sum_{i=1}^{n} (\boldsymbol{\beta}^*)^T \boldsymbol{x}_i - \sum_{i=1}^{n} y_i = 0.$$

- The first term is $\sum_{i=1}^{n} \beta_0^* = n\beta_0^*$, since $\beta_0^*$ is a constant.
- The second term is $(\boldsymbol{\beta}^*)^T \sum_{i=1}^{n} \boldsymbol{x}_i$. Since $\sum_{i=1}^{n} \boldsymbol{x}_i = \mathbf{0}$, this becomes $(\boldsymbol{\beta}^*)^T \mathbf{0} = 0$.
- The third term is $\sum_{i=1}^{n} y_i = 0$, given in the problem.

Thus:

$$n\beta_0^* + 0 - 0 = 0,$$
$$n\beta_0^* = 0.$$

Since $n > 0$, it follows that:

$$\beta_0^* = 0.$$

### Part (b): Show the equivalence of minimizers

From part (a), any global minimizer of $f(\beta_0, \boldsymbol{\beta})$ has $\beta_0^* = 0$. Thus, if $(\beta_0, \boldsymbol{\beta})$ minimizes $f$, then:

$$\beta_0 = 0,$$

and $\boldsymbol{\beta} = \boldsymbol{\beta}^*$, where $\boldsymbol{\beta}^*$ minimizes:

$$f(0, \boldsymbol{\beta}) = \sum_{i=1}^{n} \left(\boldsymbol{\beta}^T \boldsymbol{x}_i - y_i\right)^2.$$

So, the minimizers of $f$ are of the form $(0, \boldsymbol{\beta}^*)$.

Substitute the relationships $\boldsymbol{x}_i' = \boldsymbol{x}_i + \boldsymbol{q}$ and $y_i' = y_i + r$ into $\tilde{f}$:

$$\tilde{f}(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^{n} \left(\beta_0 + \boldsymbol{\beta}^T(\boldsymbol{x}_i + \boldsymbol{q}) - (y_i + r)\right)^2.$$

Simplify the expression inside the sum:

$$\beta_0 + \boldsymbol{\beta}^T(\boldsymbol{x}_i + \boldsymbol{q}) - (y_i + r) = \beta_0 + \boldsymbol{\beta}^T\boldsymbol{x}_i + \boldsymbol{\beta}^T\boldsymbol{q} - y_i - r = (\beta_0 + \boldsymbol{\beta}^T\boldsymbol{q} - r) + \boldsymbol{\beta}^T\boldsymbol{x}_i - y_i.$$

Define:

$$a = \beta_0 + \boldsymbol{\beta}^T\boldsymbol{q} - r.$$

Then:

$$\tilde{f}(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^{n} \left(a + \boldsymbol{\beta}^T \boldsymbol{x}_i - y_i\right)^2.$$

This is identical to the form of $f(a, \boldsymbol{\beta})$:

$$f(a, \boldsymbol{\beta}) = \sum_{i=1}^{n} \left(a + \boldsymbol{\beta}^T \boldsymbol{x}_i - y_i\right)^2.$$

Since $\tilde{f}(\beta_0, \boldsymbol{\beta}) = f(a, \boldsymbol{\beta})$ with $a = \beta_0 + \boldsymbol{\beta}^T \boldsymbol{q} - r$, and from part (a), $f(a, \boldsymbol{\beta})$ is minimized when $a = 0$ (because the data $(\boldsymbol{x}_i, y_i)$ is centered), the minimum of $\tilde{f}$ occurs when:

$$\beta_0 + \boldsymbol{\beta}^T \boldsymbol{q} - r = 0,$$
$$\beta_0 = r - \boldsymbol{\beta}^T \boldsymbol{q},$$

and $\boldsymbol{\beta} = \boldsymbol{\beta}^*$, where $\boldsymbol{\beta}^*$ minimizes:

$$\sum_{i=1}^{n} \left(\boldsymbol{\beta}^T \boldsymbol{x}_i - y_i\right)^2.$$

Thus, the minimizer of $\tilde{f}$ is:

$$(\beta_0, \boldsymbol{\beta}) = (r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}, \boldsymbol{\beta}^*).$$

### Verify the “if” direction

Suppose $(\beta_0, \boldsymbol{\beta})$ minimizes $f$. Then $\beta_0 = 0$ and $\boldsymbol{\beta} = \boldsymbol{\beta}^*$. Compute:

$$\beta_0 - \boldsymbol{\beta}^T \boldsymbol{q} + r = 0 - (\boldsymbol{\beta}^*)^T \boldsymbol{q} + r = r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}.$$

So:

$$(\beta_0 - \boldsymbol{\beta}^T \boldsymbol{q} + r, \boldsymbol{\beta}) = (r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}, \boldsymbol{\beta}^*),$$

which matches the minimizer of $\tilde{f}$. Thus, if $(\beta_0, \boldsymbol{\beta})$ minimizes $f$, then $(\beta_0 - \boldsymbol{\beta}^T \boldsymbol{q} + r, \boldsymbol{\beta})$ minimizes $\tilde{f}$.

### Verify the “only if” direction

Suppose $(\beta_0 - \boldsymbol{\beta}^T \boldsymbol{q} + r, \boldsymbol{\beta})$ minimizes $\tilde{f}$. Since the minimizer of $\tilde{f}$ is $(r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}, \boldsymbol{\beta}^*)$:

$$(\beta_0 - \boldsymbol{\beta}^T \boldsymbol{q} + r, \boldsymbol{\beta}) = (r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}, \boldsymbol{\beta}^*).$$

Equate components:

- $\boldsymbol{\beta} = \boldsymbol{\beta}^*$,
- $\beta_0 - \boldsymbol{\beta}^T \boldsymbol{q} + r = r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}$.

Substitute $\boldsymbol{\beta} = \boldsymbol{\beta}^*$ into the second equation:

$$\beta_0 - (\boldsymbol{\beta}^*)^T \boldsymbol{q} + r = r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}.$$

Subtract $r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}$ from both sides:

$$\beta_0 = 0.$$

Thus, $(\beta_0, \boldsymbol{\beta}) = (0, \boldsymbol{\beta}^*)$, which minimizes $f$. Hence, if $(\beta_0 - \boldsymbol{\beta}^T \boldsymbol{q} + r, \boldsymbol{\beta})$ minimizes $\tilde{f}$, then $(\beta_0, \boldsymbol{\beta})$ minimizes $f$.

### Conclusion for Part (b)

$\therefore$ It is shown that $(\beta_0, \boldsymbol{\beta})$ minimizes $f$ (i.e., $\beta_0 = 0$, $\boldsymbol{\beta} = \boldsymbol{\beta}^*$) if and only if $(\beta_0 - \boldsymbol{\beta}^T \boldsymbol{q} + r, \boldsymbol{\beta}) = (r - (\boldsymbol{\beta}^*)^T \boldsymbol{q}, \boldsymbol{\beta}^*)$ minimizes $\tilde{f}$.

## Q5

### Part a) Find all KKT-points

Rewrite the constraint in the standard form $g(x_1, x_2) \leq 0$:

$$g(x_1, x_2) = x_1 + 2x_2 + 3 \leq 0$$

The Lagrangian is:

$$\mathcal{L}(x_1, x_2, \lambda) = x_1^2 + x_2^2 + \lambda(x_1 + 2x_2 + 3)$$

where $\lambda$ is the Lagrange multiplier. The KKT conditions are:

1. **Stationarity**: $\nabla_x \mathcal{L} = 0$
2. **Primal feasibility**: $g(x_1, x_2) \leq 0$
3. **Dual feasibility**: $\lambda \geq 0$
4. **Complementary slackness**: $\lambda g(x_1, x_2) = 0$

Compute the gradients:

- $\nabla f = \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix}$
- $\nabla g = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Stationarity gives:

$$\nabla f + \lambda \nabla g = 0 \quad \Rightarrow \quad \begin{bmatrix} 2x_1 \\ 2x_2 \end{bmatrix} + \lambda \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This yields the system:

1. $2x_1 + \lambda = 0 \quad \Rightarrow \quad \lambda = -2x_1$
2. $2x_2 + 2\lambda = 0 \quad \Rightarrow \quad x_2 + \lambda = 0 \quad \Rightarrow \quad \lambda = -x_2$

Additional conditions:

- Primal feasibility: $x_1 + 2x_2 + 3 \leq 0$
- Dual feasibility: $\lambda \geq 0$
- Complementary slackness: $\lambda(x_1 + 2x_2 + 3) = 0$

From complementary slackness, either $\lambda = 0$ or $g(x_1, x_2) = x_1 + 2x_2 + 3 = 0$. Consider both cases.
**Case 1:** $\lambda = 0$

- From (1): $2x_1 = 0 \quad \Rightarrow \quad x_1 = 0$
- From (2): $2x_2 = 0 \quad \Rightarrow \quad x_2 = 0$

Check primal feasibility:

$$x_1 + 2x_2 + 3 = 0 + 2(0) + 3 = 3 \leq 0$$

This is false, so $(x_1, x_2) = (0, 0)$ is not feasible (indeed, $0 + 2 \cdot 0 = 0 > -3$). Thus, there are no KKT points when $\lambda = 0$.

**Case 2:** $\lambda > 0$
Since $\lambda > 0$, complementary slackness implies:

$$x_1 + 2x_2 + 3 = 0$$

From the stationarity conditions:

- $\lambda = -2x_1$
- $\lambda = -x_2$

Equate the two expressions for $\lambda$:

$$-2x_1 = -x_2 \quad \Rightarrow \quad x_2 = 2x_1$$

Substitute $x_2 = 2x_1$ into the constraint equation:

$$x_1 + 2(2x_1) + 3 = 0 \quad \Rightarrow \quad x_1 + 4x_1 + 3 = 0 \quad \Rightarrow \quad 5x_1 + 3 = 0 \quad \Rightarrow \quad x_1 = -\frac{3}{5}$$

Then:

$$x_2 = 2x_1 = 2\left(-\frac{3}{5}\right) = -\frac{6}{5}$$

Compute $\lambda$:

- $\lambda = -2x_1 = -2\left(-\frac{3}{5}\right) = \frac{6}{5}$
- Check: $\lambda = -x_2 = -\left(-\frac{6}{5}\right) = \frac{6}{5}$

Both match, and $\lambda = \frac{6}{5} > 0$, satisfying dual feasibility.

Verify primal feasibility:

$$x_1 + 2x_2 + 3 = -\frac{3}{5} + 2\left(-\frac{6}{5}\right) + 3 = -\frac{3}{5} - \frac{12}{5} + \frac{15}{5} = \frac{-3 - 12 + 15}{5} = 0$$

Since $0 \leq 0$, the constraint is satisfied (equality holds, consistent with $\lambda > 0$).

Thus, the only KKT point is:

$$(x_1, x_2) = \left(-\frac{3}{5}, -\frac{6}{5}\right) \text{ with } \lambda = \frac{6}{5}$$

**Part b) Determine which of these points are global minimizers**

Now there's one KKT point: $\left(-\frac{3}{5}, -\frac{6}{5}\right)$. Analyze the problem's properties:

- **Objective function**: $f(x_1, x_2) = x_1^2 + x_2^2$ is convex (its Hessian, $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$, is positive definite).
- **Constraint**: $x_1 + 2x_2 + 3 \leq 0$ is linear, hence the feasible set is convex.

Since the problem has a convex objective and a convex feasible set, the KKT conditions are necessary and sufficient for a global minimum. The point $\left(-\frac{3}{5}, -\frac{6}{5}\right)$ satisfies all KKT conditions, so it is a global minimizer.

To confirm, consider the geometric interpretation: $x_1^2 + x_2^2$ represents the squared distance from the origin. The minimizer is the point in the feasible region (below the line $x_1 + 2x_2 = -3$) closest to the origin, which occurs on the boundary due to the objective's nature. Parameterize the boundary $x_1 + 2x_2 = -3$:

Let $x_2 = t$, then $x_1 = -3 - 2t$. The objective becomes:

$$f(t) = (-3 - 2t)^2 + t^2 = 9 + 12t + 4t^2 + t^2 = 5t^2 + 12t + 9$$

Minimize $f(t)$:

$$f'(t) = 10t + 12 = 0 \quad \Rightarrow \quad t = -\frac{12}{10} = -\frac{6}{5}$$

$$x_2 = t = -\frac{6}{5}, \quad x_1 = -3 - 2\left(-\frac{6}{5}\right) = -3 + \frac{12}{5} = -\frac{15}{5} + \frac{12}{5} = -\frac{3}{5}$$

This matches our KKT point, with:

$$f\left(-\frac{3}{5}, -\frac{6}{5}\right) = \left(-\frac{3}{5}\right)^2 + \left(-\frac{6}{5}\right)^2 = \frac{9}{25} + \frac{36}{25} = \frac{45}{25} = \frac{9}{5}$$

Thus, $\left(-\frac{3}{5}, -\frac{6}{5}\right)$ is the global minimizer.

**Part c) Are there any other global minimizers?**

Since the objective $x_1^2 + x_2^2$ is **strictly convex** (the Hessian is positive definite), and the feasible set is convex, the optimization problem has at most one global minimizer. A strictly convex function achieves its minimum at a unique point within a convex set, if a minimum exists. The KKT analysis found only one point, and the alternative parameterization confirmed a unique minimum at $t = -\frac{6}{5}$.

If there were another global minimizer, it would have the same objective value, but strict convexity implies that any distinct point has a higher value, leading to a contradiction. Thus, there are no other global minimizers.